

NICOLA ZOMER

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EXPERIENCE

Research Engineer, *Marine Weather Intelligence (MWI)* C++ Python Auray / Remote (FR)

-  Sep 2024 – Present
- Developed, optimized, and maintained an isochrone routing algorithm in C++, achieving an order-of-magnitude speedup, reducing memory usage by more than 50%, and eliminating convergence failures in edge cases. Deployed via FastAPI.
 - Led the development of a state-of-the-art multi-objective optimization algorithm for weather routing, resulting in contracts with shipping companies and an internal tool that reduced routing experts' workflows from hours to minutes.
 - Integrated multiple weather models and developed a multi-model system based on location and time.

Master Thesis on Interacting AI Agents, *University of Padua* Python Padua (IT)

-  Aug 2023 – Apr 2024
- Development of a network of interacting large language model agents, used to investigate their capabilities in solving optimization tasks and replicating complex dynamics, such as the Schelling model of segregation.
 - Ran large-scale experiments on a GPU cluster provided by CloudVeneto.
 - Supervisor: Prof. Manlio de Domenico (Complex Multilayer Networks Lab).

Research Intern in Network Modeling, *Central European University* Python R Vienna (AT)

-  Apr 2023 – Jun 2023
- Application of generative network models and Bayesian inference to understand and model the behavior of platforms in the Fediverse, analyzing 4.2 million historical data points obtained by scraping. Development of a Minorize-Maximization algorithm to efficiently perform maximum likelihood estimation with ~50k parameters.
 - Supervisor: Prof. Tiago P. Peixoto (Inverse Complexity Lab).

EDUCATION

M.Sc. in Physics of Data, *University of Padua*

 Oct 2021 – April 2024  Grade: 110/110 cum laude (GPA: 29.9/30)

Relevant courses: Laboratory of Computational Physics, Management and Analysis of Physics Datasets, Machine Learning, Neural Networks and Deep Learning, Advanced Statistics for Physics Analysis, Physical Models of Living Systems

B.Sc. in Aerospace Engineering, *University of Padua*

 Oct 2018 – Sept 2021  Grade: 110/110 cum laude (GPA: 29.1/30)

Extracurricular activity: contributed to the university team *Alba CubeSat UniPD* as Mission Analysis Engineer.
Programming languages: MATLAB and Fortran.

PUBLICATIONS

Zomer, N. & De Domenico, M. Unraveling the emergence of collective behavior in networks of cognitive agents. *npj Artificial Intelligence*, 2, 36 (2026). doi:10.1038/s44387-026-00091-5

Bertè, M., Ibrahimli, R., Koopmans, L., Valgañón, P., Zomer, N. & Colombi, D. Enhancing stop location detection for incomplete urban mobility datasets. Preprint at arXiv (2024). doi:10.48550/arXiv.2407.11579

HONORS

-  **Mentee at LeadTheFuture (LTF)**
Among the few Italian students selected to be mentees for LTF, a leading mentorship non-profit organization for students in STEM, with acceptance rate below 20%.
-  **Scholarship "Mille e una lode", University of Padua**
Scholarship based on GPA for top 3% students of each degree program.
A.Y. 2018/2019, 2019/2020, 2020/2021
-  **Bronze Medal at the National Final of the Team Mathematical Olympiad**
Competed over multiple years, with bronze as the highest result.
A.Y. 2017/2018

LANGUAGES

- Italian:** Native Speaker
- English:** Full professional proficiency
- Spanish:** Limited working proficiency
- French:** Elementary proficiency